

D027

Advanced Pathopharmacological Foundations

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A 55-year-old who has smoked for 30 years and works as a long-haul truck driver has a bronchial biopsy. The normal ciliated columnar cells lining the airway have been replaced by stratified squamous cells. The patient also has well-controlled hypertension. What adaptation is this?

- A. Hypertrophy from the hypertension.
- B. Metaplasia.
- C. Hyperplasia.
- D. Cancer, because any cell change is malignant.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A pathology report describes cells that shrank, condensed their chromatin, broke into membrane-bound fragments, and were cleared without any inflammation. The tissue came from a lab with a new electron microscope. What process is this?

- A. Necrosis.
- B. Apoptosis.
- C. Atrophy.
- D. Gangrene.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A patient who works in a hospital laundry develops a red, raised rash where a nickel watch buckle touched the skin, appearing two days after wearing it. She has seasonal allergies to pollen. Which hypersensitivity type is this?

- A. Type IV, T-cell mediated delayed hypersensitivity.
- B. Type I, because she has pollen allergies.
- C. Type II, antibody against cells.
- D. Type III, immune-complex deposition.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A patient receives the wrong blood type and within minutes has fever, back pain, and dark urine as donor red cells are destroyed by antibodies. Which type is this?

- A.** Type IV, delayed.
- B.** Type III, immune complex.
- C.** Type II, cytotoxic antibody-mediated.
- D.** Type I, because it is fast.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A patient has had inflamed tissue around a retained foreign body for three months. The patient is a retired firefighter and enjoys hiking. Which cells will biopsy most likely show?

- A.** Mainly eosinophils, from an allergy.
- B.** No inflammatory cells, since the injury is old.
- C.** Mainly neutrophils.
- D.** Macrophages and lymphocytes, with possible fibrosis.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A patient with cirrhosis has an albumin of 2.1 g/dL (low), swollen legs, and ascites. He drinks two cups of coffee daily and is a retired accountant. What best explains the edema?

- A.** High plasma oncotic pressure pulling fluid out.
- B.** Increased capillary permeability from infection.
- C.** Fluid overload from coffee intake.
- D.** Low plasma oncotic pressure from low albumin lets fluid leave the vessels.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A patient with heart failure has leg edema and normal albumin. Which Starling force is most likely changed?

- A.** Lymphatic blockage from surgery.
- B.** Decreased plasma oncotic pressure.
- C.** Increased capillary hydrostatic pressure from venous congestion.
- D.** Decreased capillary hydrostatic pressure.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A patient receives 400 mg of a drug with a half-life of 6 hours. The drug is taken with breakfast and the patient weighs 70 kg. About how much remains after 18 hours, assuming no further doses?

- A. 133 mg
- B. 50 mg
- C. 100 mg
- D. 0 mg, since 18 hours is three times the half-life.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A drug level falls from 200 mg/L to 25 mg/L in 12 hours. What is the half-life?

- A. 4 hours
- B. 1.5 hours
- C. 12 hours
- D. 6 hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A patient in the ED after an opioid overdose, who arrived by ambulance from a shelter, is given naloxone and begins breathing normally within minutes. How does naloxone work?

- A. It permanently removes the opioid from the body.
- B. It is an opioid agonist that stimulates breathing.
- C. It speeds liver metabolism of the opioid.
- D. It is an opioid-receptor antagonist that displaces the opioid and blocks the receptor.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Drug X binds beta-1 receptors and raises heart rate. Drug Y binds the same receptors, has no effect when given alone, and prevents drug X from raising heart rate. What is drug Y?

- A. A beta-1 antagonist.
- B. A beta-1 agonist.
- C. A drug that speeds elimination of X.
- D. An inverse agonist that activates the opposite response.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A patient without diabetes, a schoolteacher who plays guitar, is admitted after a car crash. Her blood glucose is 190 mg/dL and heart rate 118. What best explains these findings?

- A.** Stress always lowers blood glucose.
- B.** Parasympathetic activation.
- C.** The acute stress response: sympathetic activation and cortisol raise heart rate and glucose.
- D.** Undiagnosed type 1 diabetes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A 55-year-old who has smoked for 30 years ~~and works as a long-haul truck driver~~ has a bronchial biopsy. The normal ciliated columnar cells lining the airway have been replaced by stratified squamous cells. ~~The patient also has well-controlled hypertension.~~ What adaptation is this?

Target word: name the adaptation **Decides:** One mature cell type replaced by another.

A	Camouflage	Hypertension causes cardiac hypertrophy, not airway cell change.
B	Correct	Columnar cells replaced by squamous cells is metaplasia.
C	Confusion set	Hyperplasia is more cells of the same type.
D	Absolute language	Metaplasia is reversible and not cancer, though it can progress.

2. Answer B Concept 2

A pathology report describes cells that shrank, condensed their chromatin, broke into membrane-bound fragments, and were cleared without any inflammation. ~~The tissue came from a lab with a new electron microscope.~~ What process is this?

Target word: type of cell death **Decides:** No inflammation; orderly fragmentation.

A	Inverted operation	Necrosis causes cell swelling and inflammation.
B	Correct	Programmed, orderly death without inflammation.
C	Adjacent category	Atrophy is cell shrinkage without death.
D	Confusion set	A form of necrosis in tissue.

3. Answer A Concept 3

A patient ~~who works in a hospital laundry~~ develops a red, raised rash where a nickel watch buckle touched the skin, appearing two days after wearing it. ~~She has seasonal allergies to pollen.~~ Which hypersensitivity type is this?

Target word: hypersensitivity type **Decides:** Delayed (48 h) contact reaction.

A	Correct	Contact dermatitis appears 48–72 hours after exposure.
B	Camouflage	Her pollen allergy is a separate Type I condition.
C	Adjacent category	Type II targets cell surfaces, as in transfusion reactions.
D	Confusion set	Immune-complex disease causes vasculitis or glomerulonephritis.

4. Answer C Concept 3 · transfer

A patient receives the wrong blood type and within minutes has fever, back pain, and dark urine as donor red cells are destroyed by antibodies. Which type is this?

Target word: type for antibody against cells **Decides:** Antibodies attack cell-surface antigens.

A	Inverted operation	Type IV takes days and is T-cell mediated.
B	Adjacent category	Complexes deposit in tissues; here cells are the direct target.
C	Correct	Antibodies destroy donor red cells.
D	Confusion set	Speed alone does not make it IgE-mediated.

5. Answer D Concept 4

A patient has had inflamed tissue around a retained foreign body for three months. ~~The patient is a retired firefighter and enjoys hiking.~~ Which cells will biopsy most likely show?

Target word: predominant cells **Decides:** Duration of months means chronic inflammation.

A	Adjacent category	Eosinophils suggest allergy or parasites.
B	Inverted operation	Chronic inflammation persists with active cells.
C	Confusion set	Neutrophils dominate acute inflammation.
D	Correct	Chronic inflammation features mononuclear cells.

6. Answer D Concept 5

A patient with cirrhosis has an albumin of 2.1 g/dL (low), swollen legs, and ascites. ~~He drinks two cups of coffee daily and is a retired accountant.~~ What best explains the edema?

Target word: edema mechanism **Decides:** Low albumin lowers plasma oncotic pressure.

A	Inverted operation	High oncotic pressure pulls fluid in, not out.
B	Adjacent category	No infection is described.
C	Camouflage	Coffee does not explain the edema.
D	Correct	Albumin holds fluid in vessels; when low, fluid shifts out.

7. Answer C Concept 5 · transfer

A patient with heart failure has leg edema and normal albumin. Which Starling force is most likely changed?

Target word: mechanism with normal albumin **Decides:** Venous congestion raises capillary pressure.

A	Adjacent category	No lymph obstruction is described.
B	Confusion set	Albumin is normal.
C	Correct	Back-up of blood raises pressure pushing fluid out.
D	Inverted operation	Lower pressure would reduce, not cause, edema.

8. Answer B Concept 6

A patient receives 400 mg of a drug with a half-life of 6 hours. ~~The drug is taken with breakfast and the patient weighs 70 kg.~~ About how much remains after 18 hours, assuming no further doses?

Target word: amount remaining **Decides:** 18 h = 3 half-lives.

A	Inverted operation	Divides 400 by 3 instead of halving three times.
B	Correct	400 → 200 → 100 → 50.
C	Dropped element	Stops after two half-lives.
D	Absolute language	A drug is not fully cleared until about 4–5 half-lives, and never quite zero.

9. Answer A Concept 6 · transfer

A drug level falls from 200 mg/L to 25 mg/L in 12 hours. What is the half-life?

Target word: solve for half-life **Decides:** 200 → 100 → 50 → 25 is three halvings.

A	Correct	12 h / 3 half-lives = 4 h.
B	Inverted operation	Divides 12 by the 8-fold drop instead of by 3 halvings.
C	Wrong stem number	The total time, not the half-life.
D	Confusion set	Assumes two half-lives.

10. Answer D Concept 7

A patient in the ED after an opioid overdose, ~~who arrived by ambulance from a shelter,~~ is given naloxone and begins breathing normally within minutes. How does naloxone work?

Target word: mechanism **Decides:** Blocks the opioid receptor.

A	Absolute language	Naloxone can wear off before the opioid, so effects may return.
B	Inverted operation	An agonist would add opioid effect.
C	Adjacent category	A pharmacokinetic idea; naloxone acts at the receptor.
D	Correct	Antagonists bind without activating.

11. Answer A Concept 7 · transfer

Drug X binds beta-1 receptors and raises heart rate. Drug Y binds the same receptors, has no effect when given alone, and prevents drug X from raising heart rate. What is drug Y?

Target word: classify from the effect **Decides:** Binds, does not activate, blocks.

A	Correct	Binding without activation while blocking an agonist is antagonism.
B	Inverted operation	That describes drug X.
C	Confusion set	Pharmacokinetic interaction, not receptor blocking.
D	Adjacent category	The stem says Y has no effect alone.

12. Answer C Concept 8

A patient without diabetes, ~~a schoolteacher who plays guitar,~~ is admitted after a car crash. Her blood glucose is 190 mg/dL and heart rate 118. What best explains these findings?

Target word: cause **Decides:** Acute stress response.

A	Absolute language	Stress raises glucose.
B	Inverted operation	Parasympathetic tone slows the heart.
C	Correct	Stress hormones raise glucose and heart rate.
D	Confusion set	Stress hyperglycemia is expected here without other signs.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	1	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Cellular adaptation	Hypertrophy = bigger cells; hyperplasia = more cells; atrophy = smaller cells; metaplasia = one mature cell type replaced by another; dysplasia = disordered growth that can precede cancer.	Mixing up bigger cells (hypertrophy) with more cells (hyperplasia), or calling metaplasia cancer.	A tissue under chronic stress described by size, number, or type of cells.
2	Necrosis vs. apoptosis	Necrosis is uncontrolled cell death from injury that triggers inflammation; apoptosis is programmed, orderly cell death with no inflammation.	Thinking all cell death causes inflammation.	Whether inflammation follows the cell death.
3	Hypersensitivity types	Type I IgE-mediated immediate; Type II antibody against cell-surface antigens; Type III immune-complex deposition; Type IV T-cell-mediated delayed.	Calling every allergic-sounding reaction Type I.	Timing (minutes vs. 48-72 hours) and the immune actor named.
4	Acute vs. chronic inflammation	Acute inflammation is fast and neutrophil-driven; chronic inflammation lasts weeks or more with macrophages and lymphocytes and can cause fibrosis.	Expecting neutrophils to dominate long-lasting inflammation.	Duration and the predominant white cell type.
5	Edema mechanisms	Edema forms when capillary hydrostatic pressure rises, plasma oncotic pressure falls (low albumin), capillary permeability rises, or lymph flow is blocked.	Blaming all edema on fluid overload.	Low albumin, liver or kidney disease, or burns in the history.
6	Half-life	Each half-life removes half of the remaining drug; about 4-5 half-lives reach steady state or near-complete elimination.	Thinking the full dose is gone after two half-lives.	A starting amount, a half-life, and a time elapsed.
7	Agonist vs. antagonist	An agonist binds and activates a receptor; an antagonist binds and blocks it without activating it.	Thinking an antagonist produces the opposite effect by activating something.	A drug that reverses another drug's effect at the same receptor.
8	Stress response	Acute stress activates the sympathetic system and cortisol release, raising heart rate, blood pressure, and blood glucose; chronic stress suppresses immunity.	Expecting stress to lower glucose or boost immunity long term.	A patient under physical or emotional stress with changed vital signs or labs.

Course scope (public description): Advanced Pathopharmacological Foundations provides advanced practice nurses foundational knowledge in the many pathologies encountered in practice today. Advancing from the cellular to the body system level, this course examines the pathologies of common conditions seen in healthcare today. Consideration is also given to the human affective response to alterations in health. There are no prerequisites for this course.